

Bhavesh Singh

AI Automation Engineer — Full-Stack Developer — LLMs • Agents • Next.js • PostgreSQL

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Profile

AI-focused full-stack developer with hands-on experience building autonomous AI agents, workflow automation systems, and scalable web applications. Strong foundation in backend systems, LLM integrations, databases, and modern frontend frameworks. Comfortable working across the full product lifecycle from system design to deployment.

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

Aug 2023 – Jun 2027

Bachelor of Technology in Instrumentation and Control Engineering

CGPA: 8.32 / 10

Technical Skills

Programming Languages: C++, JavaScript, TypeScript, Python

AI & LLM Systems: LLM Agents, Tool Calling, Prompt Engineering, RAG Pipelines, Embeddings, Vector Search, Memory Systems

AI Automation: n8n (advanced workflows), Webhooks, Cron Jobs, Event-driven Automation, API Orchestration

Frontend: React.js, Next.js (App Router), Tailwind CSS, Shadcn UI, Responsive UI Design

Backend: Node.js, Express.js, REST APIs, Authentication (JWT), Role-Based Access Control

Databases: PostgreSQL, Prisma ORM, MongoDB, Schema Design, Indexing, Relations

Dev Tools & Platforms: Git, GitHub, Postman, Docker (basic), Stripe API, Cal.com API, Vercel

Projects

Autonomous Voice Agent & Memory System | *Paid Client Project*

Jan 2026 – Present

- GitHub Repository — Complete implementation with workflows and custom scripts.
- Designed and deployed an autonomous AI voice agent using **n8n** to orchestrate OpenAI LLMs, ElevenLabs voice synthesis, and CRM systems.
- Implemented **Retrieval-Augmented Generation (RAG)** using Pinecone vector database for long-term conversational memory and context recall.
- Built agent logic for intent detection, memory injection, and automated callback execution without human intervention.
- Wrote custom JavaScript logic inside workflow nodes for conditional routing, state management, and payload transformations.
- Integrated scheduling and availability handling using Cal.com APIs for fully automated booking flows.

Blaezi | *Productivity & Planning Platform*

Dec 2025 – Present

- Developed a modular full-stack productivity platform using **React** and **Next.js** with scalable component architecture.
- Designed backend APIs and database schemas to support multiple feature modules and future extensibility.
- Integrated external APIs (GitHub, scheduling tools) to synchronize data across multiple services.

Tomato | *Full-Stack Food Delivery Platform*

Feb 2025 – Mar 2025

- Built a complete food delivery web application with user authentication, cart management, and order lifecycle handling.
- Implemented secure payment workflows using **Stripe**, including transaction verification and failure recovery.
- Designed role-based authentication using JWT, bcrypt, and Express middleware.
- Managed global application state and multi-user sessions using React Context API.

Positions of Responsibility

IOTA, NIT Jalandhar | *Team Leader*

Jul 2024 – Jun 2025

- Led frontend development for multiple student projects and mentored junior developers on React and system design fundamentals.

Achievements

- Solved **300+ DSA problems** across LeetCode, Codeforces, and GeeksforGeeks.
- Awarded **Bronze Medal (twice)** in Inter-Branch Football Tournament.